# 01 — Probability Foundations and Itô Calculus

**Goal of this note.** Build exactly the stochastic-calculus toolkit that every later note uses: the Itô integral, Itô's lemma (for diffusions *and* Poisson jumps), and the **generator** of a process. The generator is the single most important object here: in note 02 it turns expectations into PDEs, and in note 04 it becomes the Hamiltonian of the HJB equation.

**Prerequisites.** Measure-theoretic probability (σ-algebras, conditional expectation) at the level of a first graduate course. Nothing about stochastic processes is assumed.

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## Notation

| Symbol | Meaning | Type |
|---|---|---|
| $(\Omega, \mathcal{F}, \mathbb{P})$ | probability space: outcomes, events, probability measure | — |
| $\mathcal{F}_t$ | information available at time $t$ (filtration) | σ-algebra |
| $\mathbb{E}[\cdot]$, $\mathbb{E}[\cdot \mid \mathcal{F}_t]$ | expectation, conditional expectation given time-$t$ information | — |
| $W_t$ | standard Brownian motion | real-valued process |
| $\Delta t$, $\Delta W$ | small time step, Brownian increment over it | real |
| $H_t$ | integrand of a stochastic integral | adapted process |
| $X_t$ | an Itô process (solution of an SDE) | real-valued process |
| $b$, $\sigma$ | drift and volatility of $X$ | functions |
| $f(t,x)$ | smooth test function, $f \in C^{1,2}$ (once in $t$, twice in $x$) | function |
| $\partial_t f$, $\partial_x f$, $\partial_{xx} f$ | partial derivatives | functions |
| $N_t$ | Poisson process with intensity $\lambda$ | integer-valued process |
| $\lambda$ | jump intensity (expected jumps per unit time) | $> 0$ |
| $\eta$ | jump size of $X$ when $N$ jumps | real |
| $\tilde{N}_t = N_t - \lambda t$ | compensated Poisson process | process |
| $\mathcal{L}$ | generator of $X$ | operator on functions |
| $[X]_t$ | quadratic variation of $X$ up to $t$ | process |

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# 1. Setup & definitions

**Definition 1.1 (Filtration).** A family $\{\mathcal{F}_t\}_{t\geq 0}$ of σ-algebras with $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ for $s \leq t$. Read $\mathcal{F}_t$ as "everything that is known at time $t$." It only grows: information is never forgotten.

**Definition 1.2 (Adapted process).** A process $H_t$ is adapted if, for each $t$, $H_t$ is $\mathcal{F}_t$-measurable. In words: the value of $H_t$ can be computed from what is known at time $t$. A trading strategy must be adapted — you cannot trade on tomorrow's price.

**Definition 1.3 (Martingale).** An adapted process $M_t$ with $\mathbb{E}|M_t| < \infty$ and

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s \qquad (s \leq t). \tag{1.1}$$

A martingale is a "fair game": the best forecast of its future value is its current value. In particular $\mathbb{E}[M_t] = M_0$.

**Definition 1.4 (Brownian motion).** A process $W_t$ adapted to $\mathcal{F}_t$ such that 1. $W_0 = 0$ and $t \mapsto W_t$ is continuous; 2. for $s < t$, the increment $W_t - W_s$ is independent of $\mathcal{F}_s$; 3. $W_t - W_s \sim \mathcal{N}(0,\, t-s)$.

Two facts we will use constantly: $\mathbb{E}[\Delta W] = 0$ and $\mathbb{E}[(\Delta W)^2] = \Delta t$, so $\Delta W$ is typically of size $\sqrt{\Delta t}$, which is **much larger** than $\Delta t$ when $\Delta t$ is small.

> **Intuition:** this $\sqrt{\Delta t}$ scaling is the whole reason stochastic calculus differs from ordinary calculus. A path that moves $\sqrt{\Delta t}$ in time $\Delta t$ has slope $1/\sqrt{\Delta t} \to \infty$: Brownian paths are nowhere differentiable, so $dW_t/dt$ does not exist and we cannot use the ordinary chain rule.

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# 2. Quadratic variation: why $(dW)^2 = dt$

**Roadmap.** We show that although each $(\Delta W)^2$ is random, their sum over a partition of $[0,t]$ is *deterministic* in the limit and equals $t$. This is the precise meaning of the slogan $(dW)^2 = dt$.

Partition $[0,t]$ into $n$ equal steps $\Delta t = t/n$, with increments $\Delta W_i = W_{t_{i+1}} - W_{t_i}$. Define

$$Q_n = \sum_{i=0}^{n-1} (\Delta W_i)^2. \tag{2.1}$$

**Mean.** By Definition 1.4(3), $\mathbb{E}[(\Delta W_i)^2] = \Delta t$, so $\mathbb{E}[Q_n] = n\Delta t = t$.

**Variance.** The $\Delta W_i$ are independent, so the variances add. For $Z \sim \mathcal{N}(0, \Delta t)$, $\mathbb{E}[Z^4] = 3\Delta t^2$, so $\mathrm{Var}(Z^2) = 3\Delta t^2 - \Delta t^2 = 2\Delta t^2$. Hence

$$\mathrm{Var}(Q_n) = n \cdot 2\Delta t^2 = \frac{2t^2}{n} \xrightarrow[n\to\infty]{} 0. \tag{2.2}$$

A random variable with fixed mean $t$ and variance tending to $0$ converges (in $L^2$) to the constant $t$. So

$$[W]_t := \lim_{n\to\infty} Q_n = t. \tag{2.3}$$

**Recap.** The squared increments of $W$ add up to exactly $t$. For a smooth function $g$, the same sum $\sum(\Delta g)^2 \approx \sum g'^2 \Delta t^2 \to 0$. This non-vanishing second-order term is what Itô's lemma accounts for.

**Multiplication table.** By the same argument (each term is a sum whose variance vanishes):

| $\times$ | $dt$ | $dW$ |
|---|---|---|
| $dt$ | $0$ | $0$ |
| $dW$ | $0$ | $dt$ |

For example, $\sum \Delta t\, \Delta W_i$ has mean $0$ and variance $n\Delta t^3 \to 0$, so $dt\, dW = 0$.

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# 3. The Itô integral

**Roadmap.** We want to define $\int_0^T H_t\, dW_t$ for adapted $H$. Since $W$ has no derivative, we cannot write $\int H\, \dot{W}\, dt$. Instead we use Riemann-type sums, with one essential choice: **evaluate $H$ at the left endpoint**.

**Definition 3.1 (Simple process).** $H$ is simple if $H_t = h_i$ for $t \in [t_i, t_{i+1})$, where each $h_i$ is $\mathcal{F}_{t_i}$-measurable and $\mathbb{E}[h_i^2] < \infty$. Define

$$\int_0^T H_t\, dW_t := \sum_i h_i\, (W_{t_{i+1}} - W_{t_i}). \tag{3.1}$$

> **Intuition:** read $h_i$ as a position held over $[t_i, t_{i+1})$ and $\Delta W_i$ as the price change. Then (3.1) is trading P&L. The position must be chosen at the *start* of the interval (left endpoint), using only information available then. This is exactly the non-anticipation constraint of trading.

**Proposition 3.2 (Zero mean).** $\mathbb{E}\big[\int_0^T H\, dW\big] = 0$.

*Proof.* For each term, condition on $\mathcal{F}_{t_i}$. Since $h_i$ is known at $t_i$ and $\Delta W_i$ is independent of $\mathcal{F}_{t_i}$ with mean $0$:

$$\mathbb{E}[h_i \Delta W_i] = \mathbb{E}\big[h_i\, \mathbb{E}[\Delta W_i \mid \mathcal{F}_{t_i}]\big] = \mathbb{E}[h_i \cdot 0] = 0. \qquad \square$$

**Proposition 3.3 (Itô isometry).**

$$\mathbb{E}\Big[\Big(\int_0^T H_t\, dW_t\Big)^2\Big] = \mathbb{E}\Big[\int_0^T H_t^2\, dt\Big]. \tag{3.2}$$

*Proof.* Expand the square: $\big(\sum_i h_i \Delta W_i\big)^2 = \sum_i h_i^2 \Delta W_i^2 + 2\sum_{i<j} h_i h_j \Delta W_i \Delta W_j$. - Diagonal terms: condition on $\mathcal{F}_{t_i}$: $\mathbb{E}[h_i^2 \Delta W_i^2] = \mathbb{E}[h_i^2]\, \Delta t_i$. - Cross terms ($i < j$): $h_i h_j \Delta W_i$ is known at $t_j$, and $\Delta W_j$ is independent of $\mathcal{F}_{t_j}$ with mean $0$, so the expectation is $0$.

Summing gives $\sum_i \mathbb{E}[h_i^2]\Delta t_i = \mathbb{E}\int_0^T H_t^2\, dt$. $\square$

**Extension.** (3.2) says the map $H \mapsto \int H\, dW$ is an isometry from $L^2(\Omega \times [0,T])$ (adapted) into $L^2(\Omega)$. Simple processes are dense in the adapted $L^2$ processes, so the integral extends uniquely by continuity to every adapted $H$ with $\mathbb{E}\int_0^T H_t^2\, dt < \infty$. Propositions 3.2–3.3 continue to hold,

and moreover $t \mapsto \int_0^t H\,dW$ is a martingale.

> **Caution:** if only $\int_0^T H_t^2 dt < \infty$ almost surely (no expectation), the integral still exists but is only a *local* martingale and may have nonzero mean. This matters later: verification theorems in note 04 must check integrability before setting $\mathbb{E}\int H\,dW = 0$.

### Worked example 3.4: $\int_0^T W_t\,dW_t$

**Roadmap.** Ordinary calculus would give $\frac{1}{2}W_T^2$. We show the correct answer has an extra $-\frac{1}{2}T$, which comes directly from quadratic variation.

Use the algebraic identity $a(b-a) = \frac{1}{2}(b^2-a^2) - \frac{1}{2}(b-a)^2$ with $a = W_{t_i}$, $b = W_{t_{i+1}}$:

$$\sum_i W_{t_i}\Delta W_i = \tfrac{1}{2}\sum_i \left(W_{t_{i+1}}^2 - W_{t_i}^2\right) - \tfrac{1}{2}\sum_i (\Delta W_i)^2.$$

The first sum telescopes to $\frac{1}{2}W_T^2$ (since $W_0 = 0$). The second is $\frac{1}{2}Q_n \to \frac{1}{2}T$ by (2.3). Therefore

$$\int_0^T W_t\,dW_t = \tfrac{1}{2}W_T^2 - \tfrac{1}{2}T. \tag{3.3}$$

**Sanity checks.** Mean: $\frac{1}{2}\mathbb{E}[W_T^2] - \frac{1}{2}T = 0$ ✓ (Prop. 3.2). Second moment: by (3.2), $\mathbb{E}\int_0^T W_t^2 dt = \int_0^T t\,dt = T^2/2$; directly, $\mathrm{Var}(\frac{1}{2}W_T^2) = \frac{1}{4}\cdot 2T^2 = T^2/2$ ✓.

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## 4. Itô processes and SDEs

**Definition 4.1 (Itô process).** A process of the form

$$X_t = X_0 + \int_0^t b_s\,ds + \int_0^t \sigma_s\,dW_s, \qquad \text{written} \quad dX_t = b_t\,dt + \sigma_t\,dW_t, \tag{4.1}$$

with $b, \sigma$ adapted and integrable enough for both integrals to exist. The differential form is **only shorthand** for the integral form.

**Definition 4.2 (SDE).** When the coefficients depend on the current state, $b_t = b(t, X_t)$ and $\sigma_t = \sigma(t, X_t)$, equation (4.1) is a stochastic differential equation. If $b, \sigma$ are Lipschitz in $x$ (uniformly in $t$) with linear growth, a unique strong solution exists — the proof is Picard iteration, as for ODEs, with the Itô isometry replacing the ODE estimate.

**Quadratic variation of $X$.** From the multiplication table, $(dX)^2 = \sigma^2 (dW)^2 = \sigma^2 dt$, i.e. $[X]_t = \int_0^t \sigma_s^2 ds$. The drift contributes nothing.

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# 5. Itô's lemma (continuous case)

**Theorem 5.1 (Itô's lemma, 1-d).** Let $X$ satisfy (4.1) and $f \in C^{1,2}$. Then

$$df(t, X_t) = \Big(\partial_t f + b_t\, \partial_x f + \tfrac{1}{2}\sigma_t^2\, \partial_{xx} f\Big)dt + \sigma_t\, \partial_x f\, dW_t, \tag{5.1}$$

with all derivatives evaluated at $(t, X_t)$.

**Derivation (the idea behind the proof). Roadmap.** Taylor-expand $f$ to second order, then use the multiplication table to decide which terms survive.

Over a small step, with $\Delta X = b\,\Delta t + \sigma\,\Delta W$:

$$\Delta f = \partial_t f\,\Delta t + \partial_x f\,\Delta X + \tfrac{1}{2}\partial_{tt} f\,\Delta t^2 + \partial_{tx} f\,\Delta t\,\Delta X + \tfrac{1}{2}\partial_{xx} f\,(\Delta X)^2 + \cdots$$

Now classify each term by its order: - $\Delta t^2$: order $\Delta t^2 \to$ vanishes when summed. - $\Delta t\,\Delta X$: order $\Delta t^{3/2} \to$ vanishes. - $(\Delta X)^2 = b^2\Delta t^2 + 2b\sigma\Delta t\Delta W + \sigma^2(\Delta W)^2$: only $\sigma^2(\Delta W)^2$ survives, and it sums to $\sigma^2\Delta t$ by Section 2. - Third-order terms: order $\Delta t^{3/2}$ or smaller $\to$ vanish.

Keeping only the survivors:

$$df = \partial_t f\,dt + \partial_x f\,(b\,dt + \sigma\,dW) + \tfrac{1}{2}\partial_{xx} f\,\sigma^2 dt,$$

which is (5.1) after grouping. The rigorous proof makes "sums to" precise using (2.2) and a localization argument; the structure is exactly this.

**Recap.** Itô = ordinary chain rule **plus** the correction $\frac{1}{2}\sigma^2\partial_{xx} f\,dt$. The correction is a drift: convexity of $f$ combined with noise produces a systematic trend (Jensen's inequality in differential form).

**Theorem 5.2 (Multi-dimensional).** Let $X \in \mathbb{R}^d$, $dX = b\,dt + \sigma\,dW$ with $W \in \mathbb{R}^m$ having independent components and $\sigma$ a $d \times m$ matrix. Then

$$df = \Big(\partial_t f + b \cdot \nabla f + \tfrac{1}{2}\operatorname{tr}\big(\sigma\sigma^\top \nabla^2 f\big)\Big)dt + (\nabla f)^\top \sigma\, dW. \tag{5.2}$$

Here $\nabla f$ is the gradient, $\nabla^2 f$ the Hessian, and tr the trace. It follows from the same Taylor argument with $dW^i\,dW^j = \delta_{ij}\,dt$, so $dX^i\,dX^j = (\sigma\sigma^\top)_{ij}dt$.

**Product rule (special case).** For two Itô processes, take $f(x, y) = xy$ in (5.2):

$$d(X_t Y_t) = X_t\,dY_t + Y_t\,dX_t + dX_t\,dY_t. \tag{5.3}$$

The last term (the *covariation*) is the new piece compared with ordinary calculus.

### Worked example 5.3: geometric Brownian motion

Let $dS = \mu S\,dt + \sigma S\,dW$ with constants $\mu, \sigma$. Apply (5.1) to $f(x) = \ln x$, using $\partial_x f = 1/x$ and $\partial_{xx} f = -1/x^2$:

$$d\ln S = \frac{1}{S}(\mu S\,dt + \sigma S\,dW) + \frac{1}{2}\Big(-\frac{1}{S^2}\Big)\sigma^2 S^2 dt = \Big(\mu - \tfrac{1}{2}\sigma^2\Big)dt + \sigma\,dW.$$

The right side has constant coefficients, so integrate directly:

$$S_T = S_0 \exp\Big(\big(\mu - \tfrac{1}{2}\sigma^2\big)T + \sigma W_T\Big). \tag{5.4}$$

Check: $\mathbb{E}[e^{\sigma W_T}] = e^{\sigma^2 T/2}$ (Gaussian MGF), so $\mathbb{E}[S_T] = S_0 e^{\mu T}$ ✓.

### Worked example 5.4: CARA utility of a stock position (preview of market making)

This example is a direct preview of the Avellaneda–Stoikov reservation price (note 12).

**Setup.** Mid price $dS_t = \sigma\, dW_t$ (arithmetic BM, as in Avellaneda–Stoikov). You hold a fixed inventory of $q$ units and cash $x$, with no trading. Your terminal wealth is $x + qS_T$. Your utility is CARA, $U(w) = -e^{-\gamma w}$, with risk aversion $\gamma > 0$.

**Compute the expected utility.** $S_T = s + \sigma W_T \sim \mathcal{N}(s, \sigma^2 T)$, so $x + qS_T \sim \mathcal{N}\big(x + qs,\ q^2\sigma^2 T\big)$. For $Z \sim \mathcal{N}(m, v)$, $\mathbb{E}[e^{-\gamma Z}] = e^{-\gamma m + \gamma^2 v/2}$. Hence

$$\mathbb{E}\big[-e^{-\gamma(x+qS_T)}\big] = -\exp\Big(-\gamma\big(x + qs - \tfrac{1}{2}\gamma q^2\sigma^2 T\big)\Big). \tag{5.5}$$

**Certainty equivalent.** The term in the inner bracket is the sure wealth you would accept instead of the risky position:

$$\mathrm{CE} = x + qs - \tfrac{1}{2}\gamma q^2\sigma^2 T. \tag{5.6}$$

Inventory risk costs $\frac{1}{2}\gamma q^2\sigma^2 T$, which is **quadratic in** $q$.

**Reservation price.** How much is one more unit worth to you? The change in CE from $q$ to $q+1$ is

$$\mathrm{CE}(q+1) - \mathrm{CE}(q) = s - \tfrac{1}{2}\gamma\sigma^2 T\big((q+1)^2 - q^2\big) = s - \gamma\sigma^2 T\big(q + \tfrac{1}{2}\big).$$

Up to the $\frac{1}{2}$ (a buy/sell averaging detail treated in note 12), this is the Avellaneda–Stoikov reservation price $r = s - q\gamma\sigma^2 T$.

**Numbers.** $s = 100$, $\sigma = 2$, $T = 1$, $\gamma = 0.1$, $q = 3$, $x = 0$: - Inventory penalty: $\frac{1}{2}\cdot 0.1 \cdot 9 \cdot 4 \cdot 1 = 1.8$, so $\mathrm{CE} = 300 - 1.8 = 298.2$. - Value of a 4th unit: $100 - 0.1\cdot 4\cdot 1\cdot 3.5 = 98.6$.

> **Intuition:** a long market maker values extra inventory *below* mid, because each extra unit adds more risk than the last (risk is quadratic in $q$). That is why a long market maker skews quotes downward.

We could also get (5.5) from Itô's lemma: applying (5.1) to $f(t,S) = -\exp(-\gamma(x+qS) + \frac{1}{2}\gamma^2 q^2\sigma^2(T-t))$ shows the $dt$-term is exactly zero, so $f(t, S_t)$ is a martingale. That "choose $f$ so the drift vanishes" step is the Feynman–Kac idea of note 02.

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## 6. Poisson processes and Itô's lemma with jumps

Market-making fills are discrete events: they are modeled with Poisson processes, not Brownian motion. The HJB of note 12 has **jump terms**, which come from this section.

**Definition 6.1 (Poisson process).** A counting process $N_t$ with $N_0 = 0$, independent increments, and $N_t - N_s \sim \text{Poisson}(\lambda(t-s))$. Equivalently, on a small interval:

$$\mathbb{P}(\Delta N = 1) = \lambda \Delta t + o(\Delta t), \quad \mathbb{P}(\Delta N = 0) = 1 - \lambda \Delta t + o(\Delta t), \quad \mathbb{P}(\Delta N \geq 2) = o(\Delta t). \tag{6.1}$$

So $\mathbb{E}[\Delta N] = \lambda \Delta t$: the intensity $\lambda$ is the expected number of jumps per unit time.

**Compensated process.** $\tilde{N}_t := N_t - \lambda t$ is a martingale: its increment has mean $\lambda \Delta t - \lambda \Delta t = 0$ and is independent of the past. Writing $dN = \lambda\, dt + d\tilde{N}$ separates the predictable part (rate $\lambda$) from pure noise.

**Multiplication rule for jumps.** $\Delta N \in \{0, 1\}$ on a small interval, so $(\Delta N)^2 = \Delta N$, i.e. $(dN)^2 = dN$. Unlike $(dW)^2 = dt$, this is **not small**: a jump is an $O(1)$ change, so a Taylor expansion would need all orders. That is why the jump part of Itô's lemma is written as a full difference instead.

**Theorem 6.2 (Itô's lemma with jumps).** Let

$$dX_t = b_t\, dt + \sigma_t\, dW_t + \eta\, dN_t, \tag{6.2}$$

so $X$ diffuses and, at each jump time of $N$, jumps by $\eta$. For $f \in C^{1,2}$,

$$df(t, X_t) = \Big(\partial_t f + b\, \partial_x f + \tfrac{1}{2}\sigma^2 \partial_{xx} f\Big) dt + \sigma\, \partial_x f\, dW_t + \big[f(t, X_{t-} + \eta) - f(t, X_{t-})\big]\, dN_t. \tag{6.3}$$

Here $X_{t-}$ is the value just before a possible jump at $t$.

**Derivation.** Between jumps, $X$ is an Itô process and (5.1) applies. At a jump time, $X$ moves from $X_{t-}$ to $X_{t-} + \eta$, so $f$ changes by exactly $f(X_{t-} + \eta) - f(X_{t-})$, no Taylor expansion needed. Adding the two contributions gives (6.3).

**Expected change.** Substitute $dN = \lambda\, dt + d\tilde{N}$ into (6.3). The $dW$ and $d\tilde{N}$ terms are martingale increments (mean zero). What remains is the expected rate of change:

$$\frac{\mathbb{E}[df \mid \mathcal{F}_t]}{dt} = \partial_t f + b\, \partial_x f + \tfrac{1}{2}\sigma^2 \partial_{xx} f + \lambda\big[f(t, x + \eta) - f(t, x)\big]. \tag{6.4}$$

> **Intuition:** the jump term reads "rate of jumps × change in $f$ if a jump happens." In market making: rate of fills × change in value if filled. With fill rate $\lambda(\delta) = Ae^{-k\delta}$, this is precisely the term $\lambda(\delta^b)\,[u(q+1) - u(q)]$ in the Avellaneda–Stoikov HJB.

## Worked example 6.3: Poisson moment generating function

Let $\phi(t) = \mathbb{E}[e^{\theta N_t}]$. Apply (6.4) to $f(x) = e^{\theta x}$ with $b = \sigma = 0$, $\eta = 1$: the jump term is $\lambda(e^{\theta(x+1)} - e^{\theta x}) = \lambda(e^\theta - 1)e^{\theta x}$. Taking expectations,

$$\phi'(t) = \lambda(e^\theta - 1)\phi(t), \quad \phi(0) = 1 \implies \phi(t) = \exp\big(\lambda t(e^\theta - 1)\big). \tag{6.5}$$

This matches the known Poisson MGF, confirming (6.4).

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# 7. The generator and Dynkin's formula

**Definition 7.1 (Generator).** For the process (6.2) with $b = b(t,x)$, $\sigma = \sigma(t,x)$, define the operator acting on $f(x)$:

$$\mathcal{L}f(x) = b\,\partial_x f + \tfrac{1}{2}\sigma^2 \partial_{xx} f + \lambda\big[f(x+\eta) - f(x)\big]. \tag{7.1}$$

It is the expected instantaneous rate of change of $f(X_t)$, given $X_t = x$: $\mathcal{L}f(x) = \lim_{h\to 0} \frac{\mathbb{E}[f(X_{t+h})|X_t=x] - f(x)}{h}$. For a pure diffusion, drop the jump term; in $d$ dimensions use (5.2).

**Theorem 7.2 (Dynkin's formula).** Integrate (6.3) from $t$ to $T$ and take expectations. If the stochastic-integral terms are true martingales (e.g. $\mathbb{E}\int \sigma^2(\partial_x f)^2 ds < \infty$ and $f$ is bounded on jumps), they have zero mean, leaving

$$\mathbb{E}\big[f(T, X_T) \mid X_t = x\big] = f(t,x) + \mathbb{E}\Big[\int_t^T (\partial_s + \mathcal{L}) f(s, X_s)\, ds \;\Big|\; X_t = x\Big]. \tag{7.2}$$

**Recap and why it matters.** Dynkin's formula is the bridge from stochastic processes to PDEs: - If $(\partial_t + \mathcal{L})f = 0$, then $f(t,x) = \mathbb{E}[f(T,X_T) \mid X_t = x]$. This is **Feynman–Kac** (note 02). - If we also choose a control to make $(\partial_t + \mathcal{L}^\alpha)v$ as large as possible, we get the **HJB equation** (note 04).

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# 8. Numerical check

Script: `code/ito_checks.py`; tests: `code/tests/test_01_ito.py` (run `python -m pytest code/tests -q`). Each test compares a closed form from this note with simulation:

| Check | Closed form | Equation |
|---|---|---|
| Quadratic variation | $Q_n \to T$, $\mathrm{Var}(Q_n) = 2T^2/n$ | (2.2)–(2.3) |
| $\int W\,dW$ | left-point sum $\to \frac{1}{2}W_T^2 - \frac{1}{2}T$ pathwise | (3.3) |
| Itô isometry | $\mathbb{E}[(\int_0^T W\,dW)^2] = T^2/2$ | (3.2) |
| GBM | $\mathbb{E}[S_T] = S_0 e^{\mu T}$ | (5.4) |
| CARA certainty equivalent | $298.2$ for the numbers in Ex. 5.4 | (5.6) |
| Poisson MGF | $\exp(\lambda t(e^\theta - 1))$ | (6.5) |

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# 9. Pitfalls

1. **Left endpoint matters.** Using the midpoint in (3.1) gives the *Stratonovich* integral, under which the ordinary chain rule holds but the zero-mean property fails. Finance uses Itô because strategies are non-anticipating.
2. **Local vs. true martingale.** $\int H\,dW$ has zero mean only under integrability. Dropping it without checking is the most common gap in HJB verification arguments.
3. $(dN)^2 = dN$**, not** $0$**.** Jumps are $O(1)$; never Taylor-expand across a jump. Use the full difference $f(x+\eta) - f(x)$.

4. **Drift correction direction.** $\frac{1}{2}\sigma^2\partial_{xx}f$ is positive for convex $f$: $\mathbb{E}[S_T]$ grows at rate $\mu$, while the median of GBM grows at $\mu - \frac{1}{2}\sigma^2$.

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## 中文小结

- 基础概念：信息流 $\mathcal{F}_t$ 代表"$t$ 时刻已知的信息"；适应过程不使用未来信息；鞅是"公平游戏"，$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$。
- 二次变差：布朗运动增量量级为 $\sqrt{\Delta t}$，平方和收敛到确定值 $t$，即 $(dW)^2 = dt$；这是随机微积分区别于普通微积分的根本原因。
- **Itô** 积分：取左端点（对应交易策略不能预知未来），均值为零，满足 Itô 等距 $\mathbb{E}[(\int H\,dW)^2] = \mathbb{E}\int H^2 dt$；例：$\int_0^T W\,dW = \frac{1}{2}W_T^2 - \frac{1}{2}T$。
- **Itô** 引理：普通链式法则加修正项 $\frac{1}{2}\sigma^2\partial_{xx}f\,dt$；带跳时增加 $[f(x+\eta) - f(x)]\,dN$，期望变化率为"跳跃强度 × 跳跃带来的变化"——这正是做市 HJB 中的成交项。
- 做市预览：CARA 效用下持有 $q$ 单位的风险成本为 $\frac{1}{2}\gamma q^2\sigma^2 T$（二次），多头时额外一单位的价值低于中间价，因此报价下移。例：$q = 3$ 时确定性等价为 298.2。
- 生成元与 **Dynkin** 公式：$\mathcal{L}$ 是 $f(X_t)$ 的期望瞬时变化率；Dynkin 公式连接期望与 PDE，是 Feynman–Kac（第 02 篇）和 HJB（第 04 篇）的出发点。